Blockchain for Digital Identity & E-Governance

CASE STUDY | CYBERSECURITY | E-GOVERNANCE | DIGITAL IDENTITY

Division : 2

Class : CS-D

Group 3

1. **Ram Gavade** (29)
2. **Pranav Gawande** (35)
3. **Pruthviraj Gawande** (36)
4. **Ayush Shirode** (75)

Guided by

Prof. Manisha More

INTRODUCTION

Introduction & What is Blockchain?

Blockchain is a secure, decentralized digital ledger that records transactions transparently and cannot be easily changed.

Blockchain is a technology that stores data in linked blocks across multiple computers, making it secure and tamper-proof.

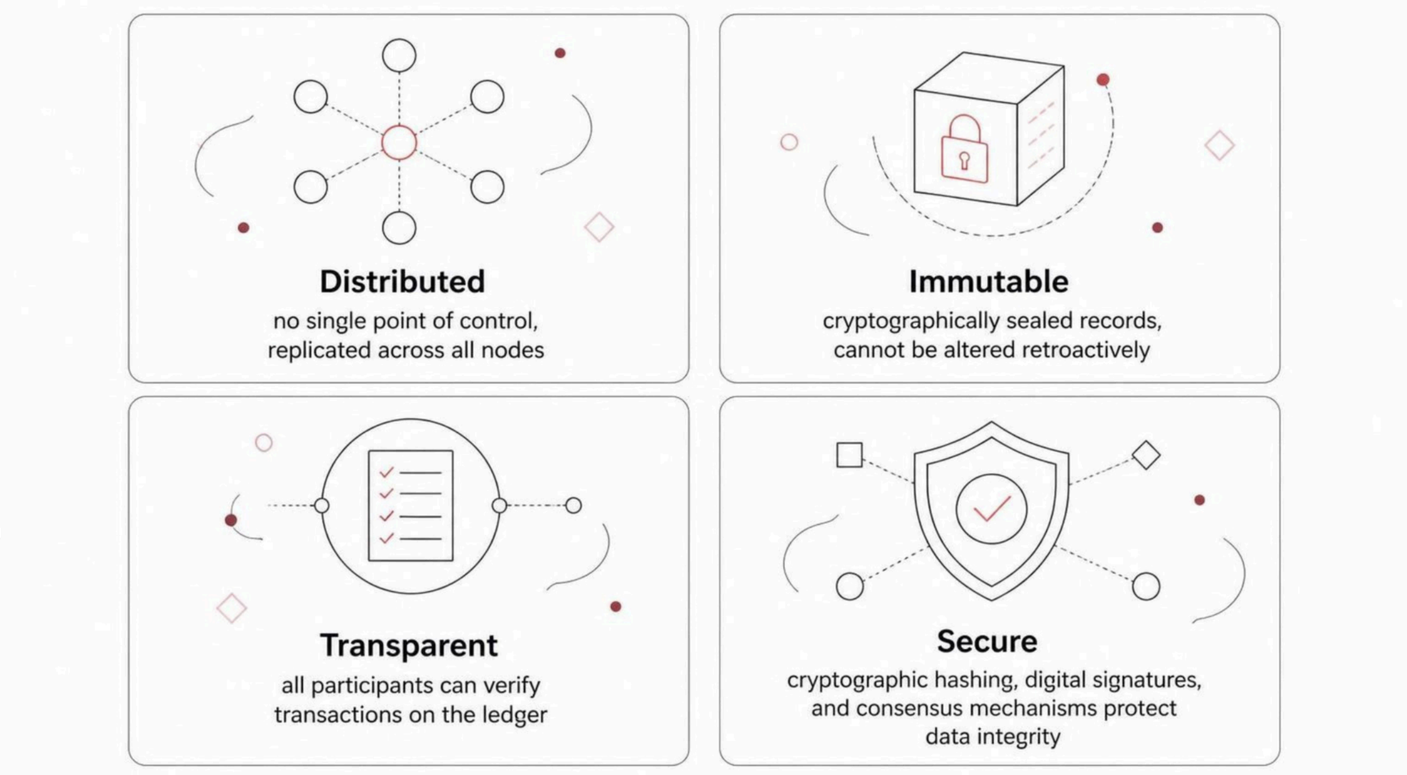
The Problem

- Fragmented identity records across agencies
- Single points of failure and data breaches
- No citizen control over personal data
- Slow, paper-heavy bureaucratic processes

The Blockchain Solution

A distributed ledger that records transactions across a peer-to-peer network without a central authority. Data is cryptographically linked in immutable blocks — once written, it cannot be altered.

- **Distributed:** No single point of control
- **Immutable:** Records cannot be altered retroactively
- **Transparent:** All participants can verify
- **Secure:** Cryptographic hashing and consensus



IDENTITY & MECHANISM

Why Blockchain for Digital Identity?

Traditional identity systems store personal data in centralized government databases — honeypots for attackers. Blockchain enables Self-Sovereign Identity (SSI), where citizens hold verifiable credentials in digital wallets and share only what is necessary.

Citizen Ownership

Individuals control their identity data via cryptographic wallets — no central authority holds all records.

Minimal Disclosure

Zero-knowledge proofs allow citizens to prove attributes (e.g., "over 18") without revealing the underlying data.

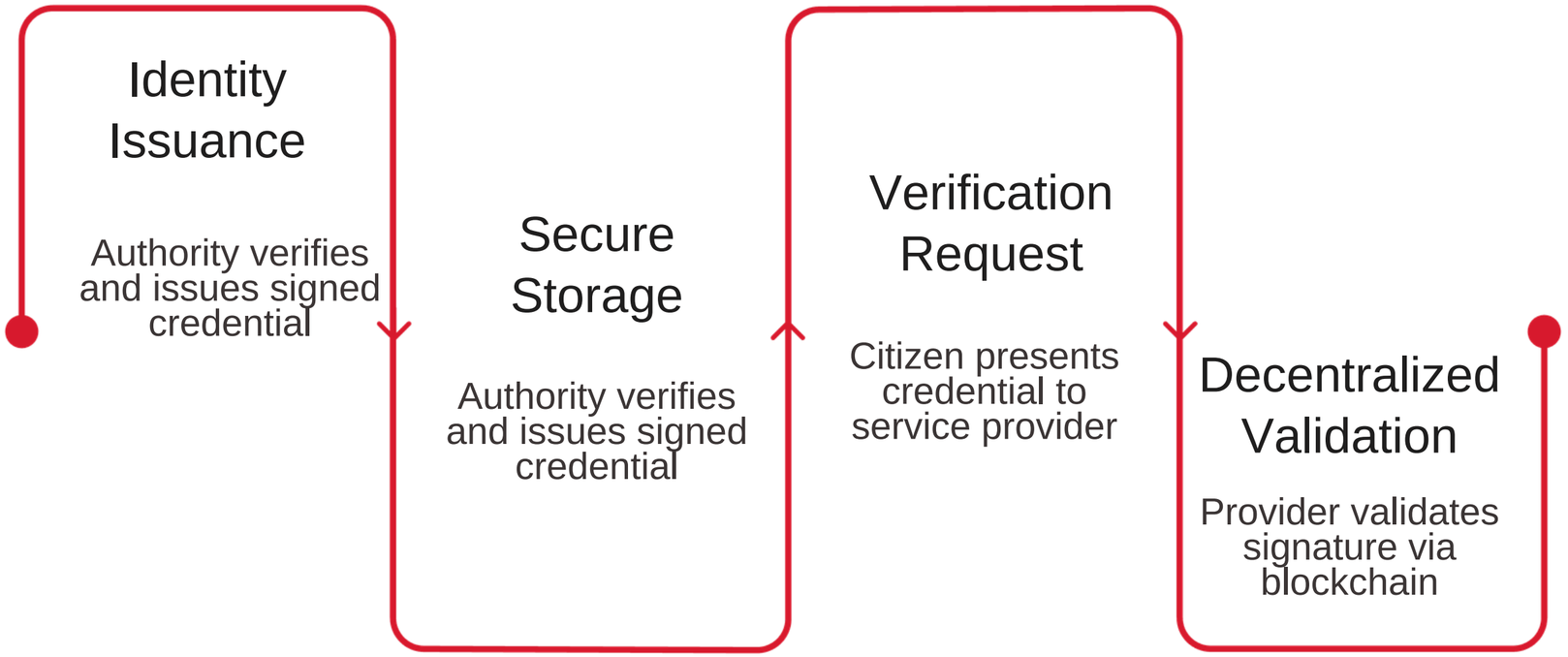
Verifiable Credentials

Issuers (governments, universities) sign credentials cryptographically. Verifiers check validity without contacting the issuer.

Cross-Agency Interoperability

A single verified identity works across all government services — tax, healthcare, voting, and licensing.

Working of Blockchain-Based Identity



IDENTITY ECOSYSTEM

Who Trusts Whom? The SSI Ecosystem

A blockchain-based identity system connects three roles — issuer, holder, and verifier.

ISSUER

Government / University

- Verifies identity
- Issues signed credential
- Can revoke credential

ISSUE →

HOLDER

Citizen / User

- Stores credentials in wallet
- Controls sharing
- Presents proof when needed

PRESENT →

VERIFIER

Government Service / Hospital / Bank

- Requests required proof
- Checks authenticity
- Grants access

Trust is established through cryptographic proofs — not by giving one organization control of the entire identity.

DATA & PRIVACY

What Goes On-Chain vs Off-Chain?

Blockchain stores what is needed for trust and verification — sensitive identity data remains private.

ON-CHAIN

Blockchain / Distributed Ledger

- ✓ **DID** : Decentralized identifier
- ✓ **Public Key** : Used to verify signatures
- ✓ **Credential Hash** : Integrity reference
- ✓ **Status / Revocation** : Is the credential valid?
- ✓ **Cryptographic Proof** : Evidence without data

VERIFY →

← proof / status

OFF-CHAIN

Citizen Wallet / Private Storage

- 🔒 **Name & Address** : Personal information
- 🔒 **Aadhaar / Passport Data** : Identity attributes
- 🔒 **Medical Records** : Sensitive health information
- 🔒 **Biometrics** : Highly sensitive identifiers
- 🔒 **Full Credentials** : Complete credential contents

Blockchain verifies the identity — it does not need to store the identity itself.

REAL-WORLD FLOW

How a Citizen Uses Blockchain Identity

Example: accessing a government service without repeatedly sharing the full identity record.

1

REQUEST

Service asks: “Are you eligible?”

→

2

WALLET

Citizen selects the required credential

→

3

PROOF

Wallet presents only necessary proof

→

4

VERIFY

Service checks signature + status

→

5

ACCESS

Valid proof → service granted

Example of selective disclosure

✓ Citizenship verified ✓ Age > 18 verified × Full address not shared × Full ID record not shared

The verifier learns that the requirement is satisfied — without necessarily receiving the underlying personal data.

APPLICATIONS

Applications of Blockchain Identity

Blockchain-based identity systems are being deployed across critical public service domains, replacing fragmented legacy infrastructure with unified, secure digital frameworks.

🗳️ Digital Voting

Tamper-proof voter registration and ballot verification. Each vote is cryptographically signed and recorded on an immutable ledger, enabling auditable elections without compromising anonymity.

Real-world: West Virginia (USA) piloted blockchain mobile voting (Voatz) for overseas military voters in 2018.

🏥 Healthcare Records

Patients control access to medical records. Doctors and hospitals request permission via smart contracts, ensuring data is shared only with authorized providers.

Real-world: Estonia's e-Health system (KSI Blockchain) secures nationwide medical records for its citizens .

📄 Document Verification

Academic certificates, property titles, and licenses issued on-chain. Employers and institutions verify authenticity instantly without contacting the original issuer.

Real-world: MIT issues digital diplomas on blockchain via the Blockcerts standard, verifiable instantly.

💼 Social Welfare

Eliminates duplicate identities in benefit distribution. Smart contracts automate eligibility checks, reducing fraud and ensuring aid reaches genuine recipients.

Real-world: UN World Food Programme's "Building Blocks" uses blockchain to deliver aid to refugees in Jordan.

ADVANTAGES

Advantages of Blockchain Identity

Blockchain-based identity systems offer measurable improvements over traditional centralized approaches across security, efficiency, and user autonomy dimensions.

Security

Cryptographic hashing and distributed consensus eliminate single points of failure. An attacker would need to compromise over 51% of network nodes simultaneously — computationally infeasible for large networks.

Privacy

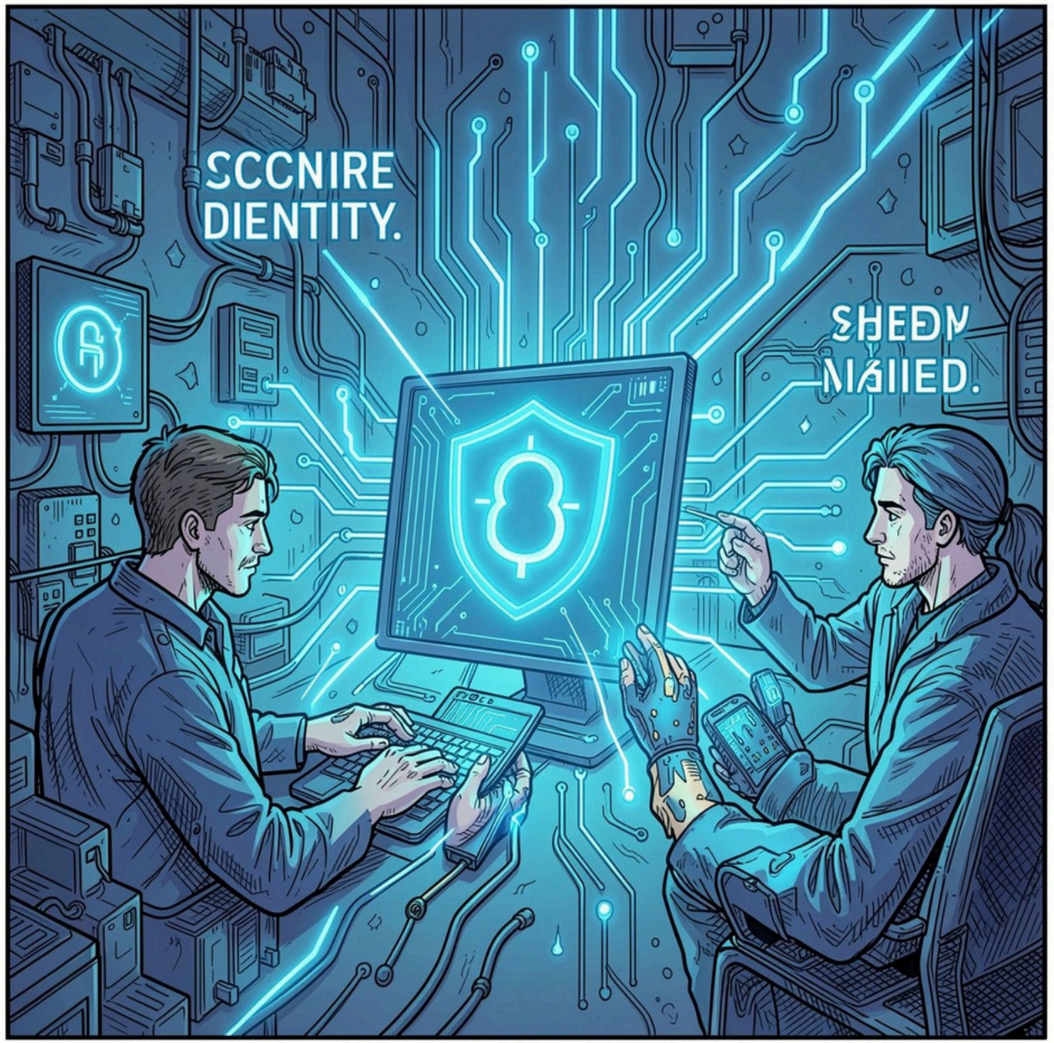
Selective disclosure and zero-knowledge proofs ensure citizens reveal only the minimum required information. Personal data never sits in a centralized honeypot vulnerable to mass breaches.

Efficiency

Automated verification via smart contracts reduces processing time from days to seconds. Cross-agency data sharing happens in real time without manual reconciliation.

Auditability

Every transaction is timestamped and permanently recorded. Regulators and citizens can audit data access logs, ensuring accountability and compliance with data protection laws.



CHALLENGES

Challenges & Estonia Case Study

Key Challenges

- **Scalability:** Public blockchains face throughput limitations under national-scale loads
- **Key Management:** Lost private keys mean permanent loss of identity — no password reset
- **Regulatory Compliance:** GDPR's "right to be forgotten" conflicts with blockchain immutability
- **Digital Divide:** Elderly and rural populations may lack devices or digital literacy
- **Interoperability:** Integrating legacy government systems with blockchain infrastructure requires significant engineering effort

Estonia: The World's First Digital Nation

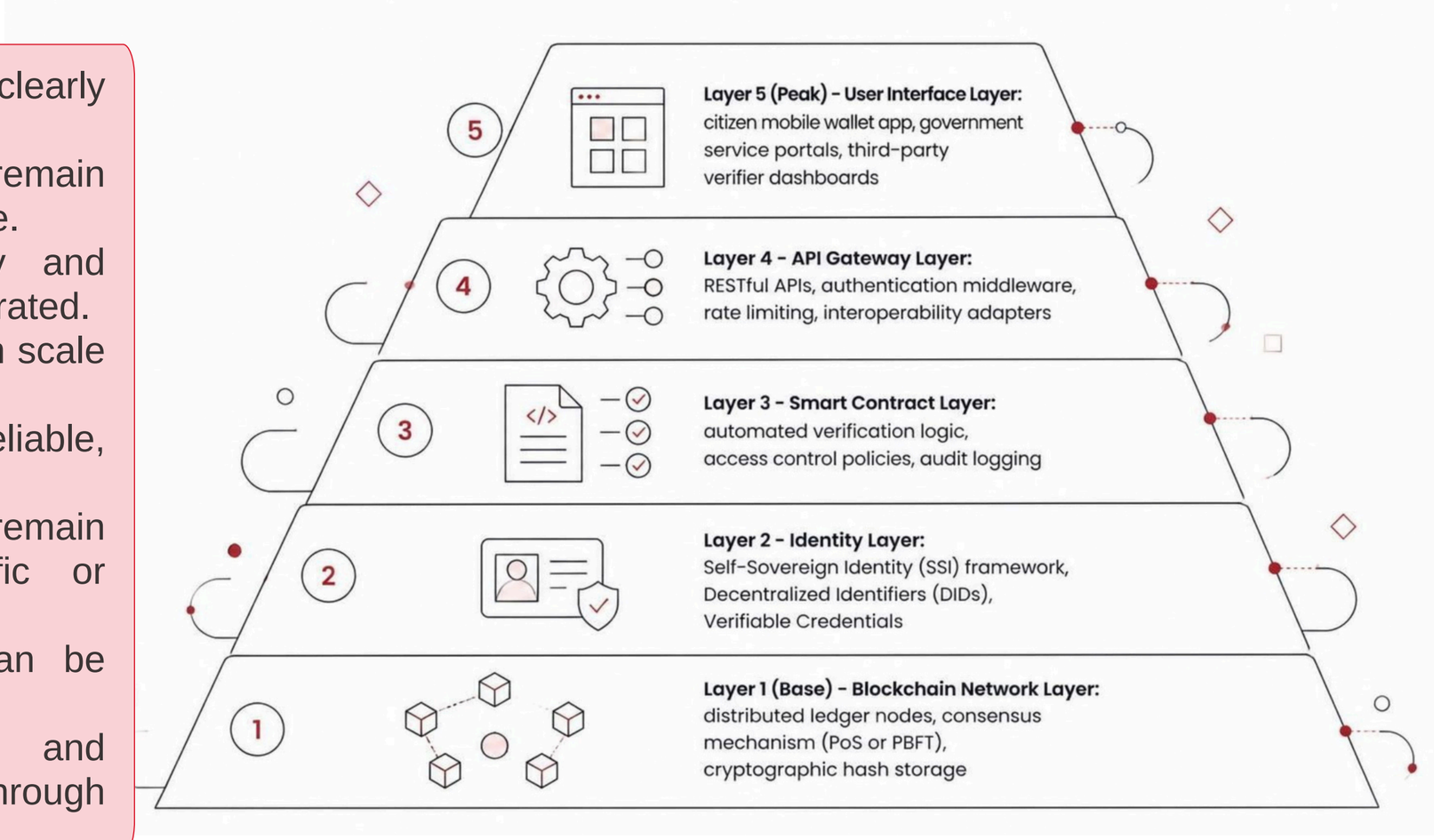
Estonia's X-Road platform, powered by KSI Blockchain (Guardtime), gives all 1.3 million citizens a state-issued digital ID. Since 2007, citizens vote, file taxes, access medical records, and sign contracts digitally — 99% of government services are online.

- **99%** of state services available online
- **2 minutes** to file annual taxes
- **30 minutes** to register a business
- **Zero** paper-based interactions required
- **KSI Blockchain** ensures data integrity across 700+ services

ARCHITECTURE

Proposed System Architecture

- Layered Architecture: Each layer has a clearly defined responsibility.
- Separation of Concerns: Components remain modular and independently manageable.
- Security Isolation: Sensitive identity and credential operations are securely separated.
- Horizontal Scalability: Infrastructure can scale to support millions of concurrent users.
- Fast Verification: Designed to provide reliable, sub-second credential verification.
- High Availability: Services should remain accessible even during high traffic or component failures.
- Modular Design: Individual layers can be upgraded.
- Secure Data Exchange: Identity and credential information is exchanged through authenticated



COMPARISON & BENEFITS

Traditional vs Blockchain Identity

Dimension	Traditional System	Blockchain System
Data Storage	Centralized government databases	Distributed across network nodes
Security Model	Perimeter defense, single point of failure	Cryptographic, no single point of failure
Data Ownership	Government or institution holds data	Citizen controls via digital wallet
Verification Speed	Hours to days (manual processes)	Seconds (automated smart contracts)
Interoperability	Siloed across agencies	Unified across all services
Audit Trail	Limited, often inaccessible	Immutable, publicly verifiable
Breach Impact	Millions of records exposed at once	No central honeypot to exploit
Cost	High (infrastructure, staffing, paper)	Lower long-term (automation)

Benefits to Society

$300B

Annual Savings

Estimated global reduction in identity fraud and administrative overhead

1.1B

People Without ID

UNHCR estimate of individuals who could gain legal identity through digital systems

99%

Service Uptime

Distributed architecture ensures near-zero downtime for critical government services

~0

Paper Documents

Fully digital workflows eliminate physical document handling and courier costs

FUTURE & CONCLUSION

Future Scope & Conclusion

2024–2025

National digital ID pilots expand across EU (EUDI Wallet), India (DigiLocker integration), and Africa. W3C Verifiable Credentials become the global standard.

2026–2027

Cross-border identity federation enables seamless travel, work, and trade. Blockchain passports and visas replace physical documents at international borders.

2028–2030

AI-powered identity risk scoring integrated with blockchain. Decentralized autonomous organizations (DAOs) use verified identity for governance participation.

2030+

Universal digital identity becomes a fundamental human right. Metaverse and IoT ecosystems rely on blockchain identity for machine-to-machine trust.

Conclusion

- Digital identity is not just about replacing a physical ID card. It is about creating a secure infrastructure through which citizens can access government services, prove who they are, and control their digital credentials.
- Blockchain can provide the foundation for security, transparency, and interoperability, while e-governance provides the real-world applications.
- The ultimate goal is a system that is secure, citizen-centric, privacy-preserving, and accessible at scale.

TAKEAWAYS

Key Takeaways

1 **Decentralization Eliminates Single Points of Failure**

Distributed ledger architecture removes the central honeypot that makes traditional identity databases attractive targets for attackers.

2 **Citizens Own Their Identity**

Self-Sovereign Identity (SSI) shifts control from institutions to individuals through cryptographic wallets and verifiable credentials.

3 **Estonia Proves It Works at National Scale**

99% of government services online, KSI Blockchain securing 700+ services — a proven blueprint for other nations.

4 **Challenges Are Engineering Problems With Solutions**

Scalability, key management, GDPR compliance, and the digital divide are active research areas with emerging solutions.

Thank You